BUSINESS REQUIREMENTS DOCUMENT

Credit Card Default Risk Analysis

A historical, data-driven study of repayment behaviour and customer default patterns

Document Type	Business Requirements Document (BRD)
Project Type	Descriptive and statistical analysis
Dataset	UCI Default of Credit Card Clients (Taiwan, 2005)
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy

1. Executive Summary

Credit card issuers face financial exposure when customers do not make required payments. This project analyses historical customer, billing and repayment information to identify patterns associated with next-month credit card default. The analysis focuses on repayment status, credit limit, outstanding bill amounts, payment amounts and derived credit-utilisation measures. Exploratory Data Analysis (EDA) and statistical hypothesis testing are used to compare default and non-default customer groups and translate the findings into practical risk-monitoring recommendations.

This is an analytical case study, not a live credit-risk decision system. It does not build a machine-learning model, assign individual risk scores, or prove that any observed factor causes default.

2. Business Background & Problem Statement

When cardholders delay or fail to repay outstanding balances, a card issuer may experience increased credit losses, collection costs and operational pressure. Risk and collections teams need to understand which customer behaviours and account characteristics are associated with subsequent payment default.

The business problem addressed is: the organisation needs a clearer, evidence-based understanding of historical repayment and financial patterns that distinguish customers who defaulted on the next month's payment from those who did not.

3. Project Objectives

- Understand the proportion of customers who defaulted and those who did not.
- Explore historical repayment-status categories and how default rates vary across them.
- Compare credit limits, bill amounts and payment amounts between default and non-default groups.
- Create summary measures for average bill amount, credit utilisation and average payment amount.
- Use appropriate statistical tests to assess group differences in credit limit, credit utilisation and average payment amount.
- Translate the analysis into practical recommendations for monitoring and collection strategy.

4. Scope

4.1 In Scope

- Load and understand the supplied UCI historical credit-card dataset.
- Review data structure, data types, missing values and potential duplicate rows.
- Conduct EDA on default distribution, repayment status, default rates by repayment status, credit limit, bill amounts and payment amounts.
- Engineer AVG_BILL_AMT, CREDIT_UTILIZATION and AVG_PAYMENT_AMT from the six historical bill/payment fields.
- Conduct three Welch's independent-samples t-tests comparing default and non-default groups.
- Document findings, limitations and business recommendations.

4.2 Out of Scope

- Building or deploying a machine-learning prediction model.
- Creating a dashboard or production reporting solution.

- Making automated credit approval, credit-limit or collections decisions.
- Establishing causal relationships between customer characteristics and default.
- Treating the historical Taiwan dataset as representative of current customers or of the Indian market without further validation.

5. Dataset Description

The project uses the UCI Default of Credit Card Clients dataset, a historical dataset from Taiwan (2005). The source file contains 30,000 customer observations and 25 columns, including a customer identifier. The project removes the identifier from the analytical feature set, leaving 24 data columns. The target field is 'default payment next month' (0 = no default, 1 = default).

Data area	Representative fields	Business meaning
Customer profile	SEX, EDUCATION, MARRIAGE, AGE	Basic customer demographic attributes
Credit facility	LIMIT_BAL	Assigned credit limit
Repayment history	PAY_0, PAY_2 to PAY_6	Historical repayment status categories
Billing history	BILL_AMT1 to BILL_AMT6	Historical statement/bill amounts
Payment history	PAY_AMT1 to PAY_AMT6	Historical payment amounts
Outcome	default payment next month	Whether the customer defaulted in the following month

Important interpretation note: PAY_0 represents the most recent repayment-status field in the dataset's historical sequence. The project should use the dataset's documented category meanings and avoid assigning calendar-month names unless the source documentation explicitly supports them.

6. Business Questions

- **Default distribution:** What proportion of customers defaulted on the next month's payment?
- **Repayment behaviour:** How are customers distributed across recent repayment-status categories, and how do observed default rates vary by category?
- **Credit limit:** Do default and non-default customers have different average credit limits?
- **Outstanding bills:** How do average bill amounts across the six historical periods compare between the two groups?
- **Payment behaviour:** How do average payment amounts across the six historical periods compare between the two groups?
- **Credit utilisation:** Do default and non-default customers have different average credit utilisation?

7. Analytical Approach & Required Activities

Data loading and preparation

Load the supplied dataset, inspect its dimensions and columns, promote the embedded header row if required by the source file format, remove the customer identifier from analysis and convert analytical fields to numeric types.

Data quality assessment

Check missing values, data types and potential duplicate rows. Record the duplicate-handling decision and its impact on the final record count.

Exploratory Data Analysis

Calculate default/non-default counts and percentages; inspect PAY_0 frequency; calculate default rates by PAY_0 category; compare credit limits, six bill-amount fields and six payment-amount fields by default status; use suitable charts to communicate distributions and differences.

Feature engineering

Calculate AVG_BILL_AMT as the row-wise mean of BILL_AMT1–BILL_AMT6. Calculate CREDIT_UTILIZATION as AVG_BILL_AMT divided by LIMIT_BAL multiplied by 100. Calculate AVG_PAYMENT_AMT as the row-wise mean of PAY_AMT1–PAY_AMT6.

Hypothesis testing

Compare default and non-default groups for credit utilisation, average payment amount and credit limit using Welch's independent-samples t-test, with a significance threshold of 0.05.

Interpretation and recommendations

Summarise statistically significant group differences, explain the business relevance, state limitations and recommend monitoring actions that are proportionate to the evidence.

8. Statistical Testing Requirements

For each test, the null hypothesis states that the two customer groups have equal population means for the selected numerical measure. The alternative hypothesis states that the means differ. The project uses a two-sided Welch's independent-samples t-test (SciPy ttest_ind with equal_var=False), which does not require the two groups to have equal variances.

Test 1: Credit utilisation vs default

- H_0: Mean credit utilisation is equal for default and non-default customers.
- H_1: Mean credit utilisation differs between default and non-default customers.

Recorded result: Observed means: 44.97% (default) vs 35.18% (non-default); t = 19.33; $p \approx 8.11 \times 10^{-82}$.

Test 2: Average payment amount vs default

- H_0: Mean average payment amount is equal for default and non-default customers.
- H_1: Mean average payment amount differs between default and non-default customers.

Recorded result: Observed means: 3,331.23 (default) vs 5,835.36 (non-default); t = -24.04; $p \approx 7.85 \times 10^{-126}$.

Test 3: Credit limit vs default

- H_0: Mean credit limit is equal for default and non-default customers.
- H_1: Mean credit limit differs between default and non-default customers.

Recorded result: Observed means: 129,984.57 (default) vs 178,084.51 (non-default); t = -29.02; $p \approx 4.79 \times 10^{-179}$.

Decision rule: compare each p-value with $\alpha = 0.05$. Since the recorded p-values are below 0.05, the null hypothesis is rejected for all three comparisons. These results indicate statistically significant differences in group means; they do not establish causation or predictive performance.

9. Functional & Analytical Requirements

ID	Requirement
FR-01	The analysis shall identify the target variable and clearly define 0 as non-default and 1 as default.

FR-02	The analysis shall report dataset dimensions, field types, missing-value count and duplicate-check results.
FR-03	The analysis shall quantify default and non-default counts and percentages.
FR-04	The analysis shall summarise recent repayment-status categories and calculate observed default rates by category.
FR-05	The analysis shall compare credit limit, historical bill amounts and historical payment amounts by default status.
FR-06	The analysis shall calculate average bill amount, credit utilisation and average payment amount using the specified six-period fields.
FR-07	The analysis shall run three Welch's independent-samples t-tests using $\alpha = 0.05$.
FR-08	The notebook shall document findings, business interpretations, limitations and recommendations alongside the analysis.

10. Business Recommendations Mapped to Findings

Finding from analysis	Suggested business response
Observed longer repayment delays coincide with higher default rates in the dataset.	Use repeated or worsening payment delays as a trigger for review and timely, customer-appropriate collection outreach.
Default customers had higher mean credit utilisation (44.97% vs 35.18%).	Monitor sustained high utilisation alongside repayment history; do not use utilisation as a standalone risk decision.
Default customers had lower mean average payment amounts (3,331.23 vs 5,835.36).	Review declining or consistently low payment patterns and consider proactive reminders or payment-support options.
Default customers had lower mean credit limits (129,984.57 vs 178,084.51).	Consider credit limit together with repayment and utilisation behaviour when reviewing customer profiles; do not infer that a lower limit causes default.
Bill amounts alone did not clearly separate default and non-default groups in the observed averages.	Avoid relying only on outstanding bill size; combine multiple behavioural and account indicators.

11. Expected Deliverables

- A reproducible Jupyter Notebook containing data preparation, EDA, feature engineering, hypothesis tests and documented findings.
- A concise set of visualisations and summary tables supporting the analysis.
- This Business Requirements Document describing the business problem, objectives, scope, analytical requirements and recommended actions.
- A GitHub repository containing the notebook, BRD/README and dataset access instructions or source attribution.

12. Success Criteria

- The notebook runs from beginning to end with the documented dataset and required libraries.
- All analytical questions in Section 6 are addressed by relevant EDA or statistical analysis.
- All three Welch's t-tests report test statistics, p-values and decisions using $\alpha = 0.05$.
- Each major finding is translated into a cautious, evidence-based business interpretation.
- The final project clearly communicates that it is a historical statistical case study, not a deployed prediction model.

13. Assumptions, Constraints & Limitations

- The source dataset is historical (Taiwan, 2005); results may not generalise to current portfolios, other countries or current underwriting/collections environments.
- The target represents default payment next month as encoded in the dataset.
- The analysis is observational. Statistically significant differences do not prove that a variable causes default.
- The data contains skewed financial variables, negative bill amounts and unusually high utilisation values; these may influence mean-based comparisons.
- Welch's t-test addresses unequal group variances but does not by itself eliminate the influence of outliers or resolve all distributional concerns.
- The project does not validate a live operational policy, model performance, fairness, regulatory compliance or financial impact.

14. Glossary

Term	Meaning
Default	Failure to make the required payment, represented by target value 1 in this dataset.
Non-default	No next-month default, represented by target value 0.
Credit limit	The credit amount assigned to a customer account.
Credit utilisation	Average bill amount divided by credit limit, expressed as a percentage.
EDA	Exploratory Data Analysis; initial examination of distributions, patterns and relationships.
Hypothesis test	A statistical procedure used to assess evidence against a stated null hypothesis.
P-value	Probability, assuming the null hypothesis and test assumptions, of observing a result at least as extreme as the one obtained.
Welch's independent t-test	A two-sample test comparing group means without assuming equal population variances.